PORJECT 1

Mini world:

Home renovation contractor

Requirement Gathering:

To figure out the requirements for my Home Renovation Contractor Management System I researched how home renovation contractors manage their customers, projects, workers, and materials I looked at the different types of information a contractor would need to keep track of when working on a renovation project I used this information to decide what the database should store and how the different parts of the system should be connected.

System Design requirements:

1. The system should keep information about each customer, such as their name, phone number, email, and address.

2. The system should keep information about each renovation project, such as the project name, description, start date, expected completion date, and status.

3. The system should keep track of which customer is connected to each renovation project.

4. The system should keep information about the workers who work for the contractor, such as their name, phone number, email, and job position.

5. The system should keep track of which workers are assigned to each renovation project.

6. **The system should keep information about the materials used for each renovation project, such as the material name, description, and quantity.**

7. The system should keep track of how much of each material is used for a renovation project.

8. The system should keep track of the cost of each renovation project.

9. The system should keep track of the location of each renovation project.

10. The system should keep track of the payments made for each renovation project.

Based on the requirements I came up with the main entities that the system needs to keep track of. These are Customer, Project, Worker, Material and Payment. Each one has different information that needs to be stored. For example the Customer keeps the customers name, phone number, email and address. The Project keeps information about the renovation project like the project name, description, start date, expected completion date, status, cost and location. The Worker keeps information about the workers like their name, phone number, email and job position. The Material keeps information about the materials used for the projects. The Payment keeps track of payments made for each project.

I also needed ways to connect the different parts of the system. A project can have multiple workers and a worker can work on multiple projects so I used Project_Worker to connect them. A project can also use multiple materials so I used Project_Material to connect the projects and materials. Project_Material also keeps track of how much of each material is used. These entities and attributes were then used when I made my ER diagrams and database